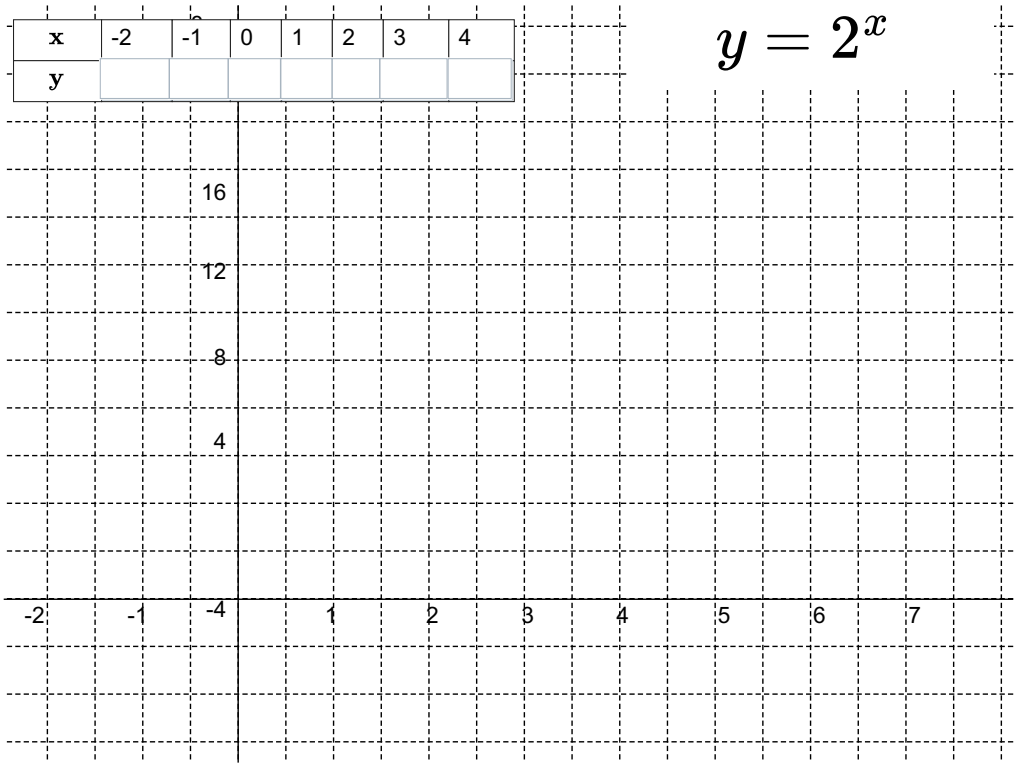
1.22 Exponentials and Logarithms

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x	-2	-1	0	1	2	3	4
y							

$y = 2^x$



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Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

1.22 Exponentials and Logarithms

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Graph Transformations

6

Sketch $y = 2^{x+3}$

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Exercise 14A (Pages 313-314)

7

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

1.22 Exponentials and Logarithms

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Logarithms

21

You know the inverse of many mathematical operations; we can undo an addition by 2 for example by subtracting 2. But is there an inverse function for an exponential function?

	Function		Inverse	
4 →	$x \times 3$	12 →		→ 4
4 →	$x + 3$	7 →		→ 4
4 →	x^2	16 →		→ 4
4 →	x^5	1024 →		→ 4
4 →	3^x	81 →		→ 4

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Interchanging between exponential and log form

22

$\log_a n$ (“said log base a of n”) is equivalent to $a^x = n$.
The log function outputs the **missing power**.

$$3^2 = 9 \iff \log_3 9 = 2$$

Here are two methods of interchanging between these forms.
Pick your favourite!

Method 1: ‘Missing Power’

- Note first the base of the log must match the base of the exponential.
- $\log_2 8$ for example asks the question “2 to **what power** gives 8?”

We can imagine inserting the output of the log just after the base. Click the button!

$\log_{\quad 2} 8 = 3$ Click to start Animation

Method 2: Do same operation to each side of equation.

Since KS3 you’re used to the idea of doing the same thing to each side of the equation that ‘undoes’ whatever you want to get rid of.

$$3x + 2 = 11$$
$$\overset{(-2)}{\quad} \quad 3x = 9 \quad {}^{(-2)}$$

“log base a” undoes “a to the power of” and vice versa, as they are inverse functions.

$$\log_2 8 = 3$$
$$(2^{\square}) \quad 8 = 2^{3} \quad {}^{(2^{\square})}$$

1.22 Exponentials and Logarithms

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Examples

23

$\log_5 25 = \mathbf{2}$ ← Think: "5 to the power of what gives you 25?"

$\log_3 81 = \mathbf{4}$

$\log_2 32 = \mathbf{5}$

$\log_{10} 1000 = \mathbf{3}$

$\log_4 1 = \mathbf{0}$

$\log_4 4 = \mathbf{1}$

$\log_2 \left(\frac{1}{2}\right) =$

$\log_3 \left(\frac{1}{27}\right) =$

$\log_2 \left(\frac{1}{16}\right) =$

$\log_a \left(a^3\right) = \mathbf{3}$

$\log_4 (-1) =$

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With Your Calculator

24

There are three buttons on your calculator for computing logs:

$\log_{\square}\square$

$\log_3 7 = \mathbf{1.7}$

$\log_5 0.3 = -\mathbf{1}$

ln

$\ln 10 = \mathbf{2.3}$

$\ln e = \mathbf{1}$

log

$\log 100 = \mathbf{2}$

$\ln$ is the "**natural log of x**", meaning "log to the base e", i.e. it the inverse of e^x.

$\ln(x) = \log_e(x)$

We will use it more extensively later this chapter.

Just like the $\sqrt{}$ symbol without a number is $\sqrt[2]{\square}$ by default, $\log$ without a base is **base 10** by default when used on your calculator (although confusingly "$\log$" can mean "$\ln$" in mathematical papers)

1.22 Exponentials and Logarithms

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Exercise 14D – Y1 Page 321

25

6 a Without using a calculator, justify why the value of $\log_2 50$ must be between 5 and 6.

b Use a calculator to find the exact value of $\log_2 50$ to 4 significant figures.

7 a Find the values of:

i $\log_2 2$ \qquad **ii** $\log_3 3$ \qquad **iii** $\log_{17} 17$

b Explain why $\log_a a$ has the same value for all positive values of a $(a \neq 1)$.

8 a Find the values of:

i $\log_2 1$ \qquad **ii** $\log_3 1$ \qquad **iii** $\log_{17} 1$

b Explain why $\log_a 1$ has the same value for all positive values of a $(a \neq 1)$.

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Extension Exercise

26

1 [MAT 2015 1J] Which is the largest of the following numbers?

A) $\frac{\sqrt{7}}{2}$ B) $\frac{5}{4}$ C) $\frac{\sqrt{10!}}{3(6!)}$

D) $\frac{\log_2 30}{\log_3 85}$ E) $\frac{1+\sqrt{6}}{3}$

Non-calculator!

2 *[MAT 2013 1F]* Three *positive* numbers a, b, c satisfy

$\log_b a = 2$

$\log_b (c-3) = 3$

$\log_a (c+5) = 2$

This information:

A) specifies a uniquely;

B) is satisfied by two values of a;

C) is satisfied by infinitely many values of a;

D) is contradictory

E)

If we take exponents of the three equations:

$$\mathbf{a=b^2,\ c-3=b^3,\ c+5=a^2}$$

Hence eliminating a and c we get

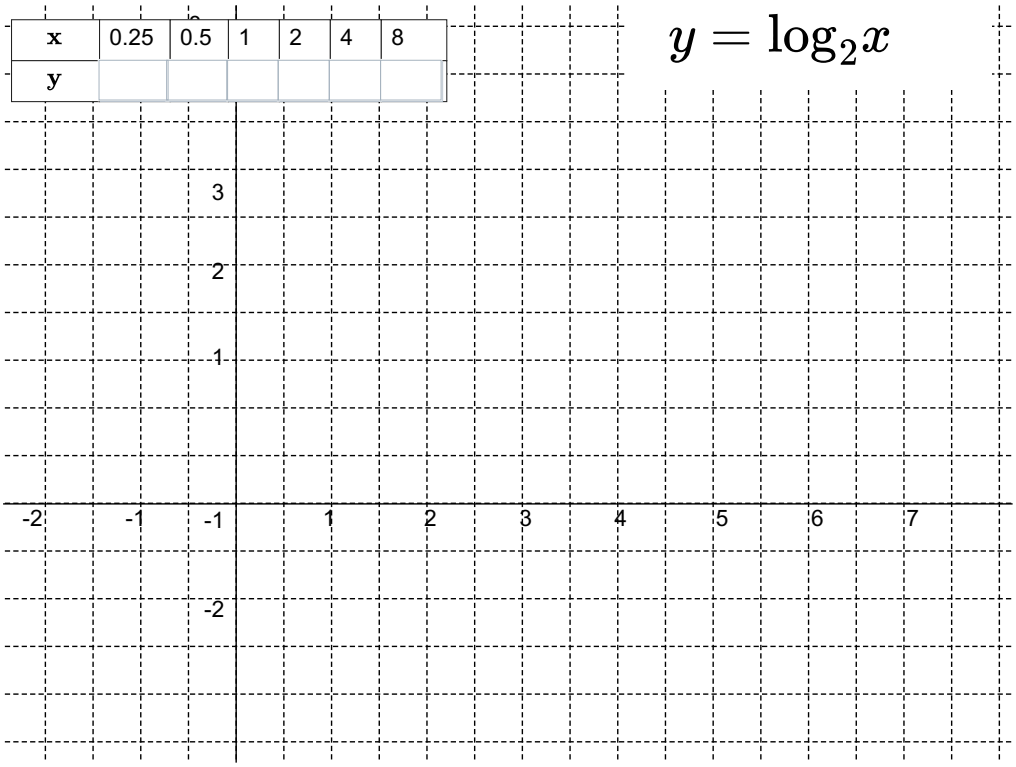
1.23 Exponentials and Logarithms

Only one positive solution to the equation ($b=2$, so that $a=4, c=11$). Answer is (a).

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x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$



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Laws of Logs

28

Three main laws:

$\log_a x + \log_a y = \log_a xy$

$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$

$\log_a \left(x^k\right) = k\log_a x$

The logs must have a consistent base.

i.e. You can move the power to the front.

Special cases:

$\log_a a = 1 \quad (a > 0, a \neq 1)$

$\log_a 1 = 0 \quad (a > 0, a \neq 1)$

$\log\left(\frac{1}{x}\right) = \log\left(x^{-1}\right) = -\log(x)$

We often try to avoid leaving fractions inside logs.
So if the answer was:
$\log_2\left(\frac{1}{3}\right)$
You should write your answer as: $-\log_2 3$
Reciprocating the input negates the output.

Not in syllabus (but in MAT/PAT):

$$\log_a b = \frac{\log_c b}{\log_c a}$$

This is known as **changing the base**.
So to get $\log_2 9$ in terms of log base 3:
$log_2 9 = \frac{log_3 9}{log_3 2} = \frac{2}{log_3 2}$

1.22 Exponentials and Logarithms

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Examples

29

Write as a single logarithm:
a. $\log_3 6 + \log_3 7$
b. $\log_2 15 - \log_2 3$
c. $2\log_5 3 + 3\log_5 2$
d. $\log_{10} 3 - 4\log_{10}\left(\frac{1}{2}\right)$

Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$
a. $\log_a(x^2yz^3)$
b. $\log_a\left(\frac{x}{y^3}\right)$
c. $\log_a\left(\frac{x\sqrt{y}}{z}\right)$
d. $\log_a\left(\frac{x}{a^4}\right)$

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Anti Laws

30

These are **NOT LAWS OF LOGS**, but are mistakes students often make:

$$\log_a(b+c) = \log_a b + \log_a c$$

× FAIL

$$(\log_2 x)^3 = 3\log_2 x$$

× FAIL

1.22 Exponentials and Logarithms

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Solving Equations with Logs

31

Solve the equation $\log_{10}4 + 2\log_{10}x = 2$

This is a very common type of exam question.
The strategy is to **combine the logs into one** and isolate on one side.

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Test Your Understanding

32

Edexcel C2 Jan 2013 Q6

Given that $2\log_2(x+15) - \log_2 x = 6$,

(a) show that $x^2 - 34x + 225 = 0$. \hfill (5)

(b) Hence, or otherwise, solve the equation $2\log_2(x+15) - \log_2 x = 6$. \hfill (2)

a

b

1.22 Exponentials and Logarithms

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Exercise 14E (323-324)

33

5 a Given that $\log_3(x+1) = 1 + 2\log_3(x-1)$, show that $3x^2 - 7x + 2 = 0$.

b Hence, or otherwise, solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

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Exercise 14E (323-324)

34

6 Given that a and b are positive constants, and that $a > b$, solve the simultaneous equations

$a + b = 13$

$\log_6 a + \log_6 b = 2$

Challenge:
By Writing $\log_a x = m$ and $\log_a y = n$
Prove that
$\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$

1.22 Exponentials and Logarithms

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Extension Exercises

35

These are all strictly **non-calculator**!

1 [AEA 2010 Q1b] Solve the equation

$$\log_3(x-7) - \tfrac{1}{2}\log_3 x = 1 - \log_3 2$$

$\mathbf{x = \frac{49}{}}$

2 na, who is confused about the rules of logarithms, states that

$(\log_3 p)^2 = \log_3(p^2)$

$\log_3(p+q) = \log_3 p + \log_3 q$

However, there is a value for p and a value for q for which both statements are correct. Find their values.

First equation:

3 [MAT 2007 1I] Given that a and b are positive and

$$4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$$

what is the greatest possible value of a?

4 [MAT 2002 1F] Observe that $2^3 = 8$, $2^5 = 32$, $3^2 = 9$ and $3^3 = 27$. From these facts, we can deduce that $\log_2 3$ is:

A) between $1\frac{1}{3}$ and $1\frac{1}{2}$

B) between $1\frac{1}{2}$ and $1\frac{2}{3}$

C) between $1\frac{2}{3}$ and 2

D) none of the above

$\mathbf{\log_2 3 > \frac{3}{2}}$

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Solving equations with exponential terms

36

Solve $3^x = 20$

Solve $5^{4x-1} = 61$

1.22 Exponentials and Logarithms

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Solving equations with exponential terms

37

Solve $3^x = 2^{x+1}$

Why can we not apply quite the same strategy here?

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Test Your Understanding

38

1 Solve $3^{2x-1} = 5$, giving your answer to 3dp.

3 Solve $2^x 3^{x+1} = 5$, giving your answer in exact form.

2 Solve $3^{x+1} = 4^{x-1}$, giving your answer to 3dp.

1.22 Exponentials and Logarithms

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Exercise 14F – Y1 Page 325

39

3 Solve the following equations, giving your answers to 3 significant figures where appropriate.

a $3^{x+1} = 2000$ \hfill (2 mar

b $\log_5(x-3) = -1$ \hfill (2 mar

4 a Sketch the graph of $y = 4^x$, stating the coordinates of any points where the graph crosses the axes.

b Solve the equation $4^{2x} - 10(4^x) + 16 = 0$.

1 **EXTENSION** [MAT 2011 1H] How many *positive* values x which satisfy the equation:

$$x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + \log_{0.5} 0.25$$

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Very Challenging Extension!

40

2 [MAT 2013 1J] For a real number x we denote by $[x]$ the largest integer less than or equal to x. Let n be a natural number. The integral

$$\int_0^n [2^x] dx$$

equals:

(A) $\log_2((2^n-1)!)$

(B) $n2^n - \log_2((2^n)!)$

(C) $n2^n$

(D) $\log_2((2^n)!)$